Kunal Sen

FPGA Design Engineer

RESUME_1017

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Total Experience: 12 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 12 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Verilog, FPGA, Timing Closure, SPI, Hardware Simulation, VHDL, AXI4, Intel Quartus, UART, Xilinx Vivado, DSP, Tcl

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | SparrowCloud Labs | Jan 2022 - Present

- Lowered end-to-end data-path latency by 24% using pipelined logic
- Verified 259 protocol corner cases with repeatable simulation suites
- Reduced FPGA resource utilization by 27% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | Foresight Industrial | Jan 2018 - Dec 2021

- Reduced FPGA resource utilization by 27% through arithmetic sharing
- Improved maximum operating frequency by 21% after timing-path optimization

Associate FPGA Design Engineer | ZenithTrail Technologies | Jul 2014 - Dec 2017

- Lowered end-to-end data-path latency by 28% using pipelined logic
- Verified 800 protocol corner cases with repeatable simulation suites

PROJECTS

Camera Data Bridge | Timing Closure, VHDL, Hardware Simulation, Verilog

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | UART, Verilog, Intel Quartus, SPI

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Shivanta College of Technology, 2014
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Digital Signal Processing Basics (2018)